

Input Sourcing, Trade Liberalization, and Diversifying Mergers

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Abstract

We show that a firm’s expertise in managing specific inputs—its core competence—is a key driver of diversifying mergers. While prior work shows that such capabilities drive organic, within-firm product expansion, we demonstrate that they also motivate firms’ inorganic growth through the acquisition of external firms. Using China’s 2001 WTO accession as a natural experiment, we find that a reduction in tariffs on common inputs of two industries significantly increases the likelihood of a diversifying merger between these two industries. This effect is more salient for differentiated and innovation-intensive inputs, which reveals the nature of the mechanism: the valuable capability is not about sourcing generic inputs, but about mastering complex, knowledge-intensive ones. The main result holds in a global sample covering 71 countries. Our findings extend the core competence theory by highlighting how input-related capabilities shape firm boundaries across industries.

Keywords: Trade Liberalization; Firm Capabilities; Core Competence; Mergers and Acquisitions; Firm Diversification

JEL Classification: F13, G34, L25, D22.

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1 Introduction

The past few decades have witnessed major reductions in trade barriers through international agreements such as the formation of NAFTA and China’s WTO accession. Extensive research has documented how trade liberalization enhances firms’ input-sourcing capabilities that boost firm performance, in particular in developing economies, where domestic firms gain access to previously restricted foreign intermediate inputs.¹

In parallel to this trade literature, research on industrial organization and corporate strategy shows that firms strategically use their unique resources and capabilities when entering new industries. The resource-based view of the firm argues that firms derive competitive advantages (or core competences) from unique, valuable, and difficult-to-imitate resources or capabilities (Wernerfelt, 1984; Prahalad and Hamel, 1990).² Among these core competences that shape firm boundaries, we highlight input capability—a firm’s accumulated knowledge and expertise in sourcing, processing, and utilizing specific inputs. Penrose (1955) pioneers the concept that firms can extend beyond single-industry constraints through diversification. Boehm et al. (2022) show that firms are more likely to leverage this capability for organic growth, starting to produce new products internally when those products share similar input requirements as their existing portfolios.

Inspired by the resource-based view of the firm, we ask a conceptually distinct question: can the input capability drive changes in the boundaries of the firm through inorganic expansion? Specifically, we investigate whether trade liberalization, by making critical inputs cheaper and more accessible, encourages firms to acquire other firms in seemingly unrelated industries that rely on a similar set of inputs. This allows us to move beyond what firms *do* to understand what changes firms’ boundaries, providing a new lens on the drivers of diversifying mergers and acquisitions (M&As).³ Our conjecture is that the answer to the above question is yes. This input-driven diversification is

¹For example, input trade liberalization has been documented to improve firm productivity (Amiti and Konings, 2007), increase product scope and quality (Goldberg et al., 2010), change innovation (Liu and Qiu, 2016), and facilitate firms’ exports (Bas and Strauss-Kahn, 2015; Fan et al., 2015).

²The combined Google Scholar citation count for Wernerfelt (1984); Prahalad and Hamel (1990); Barney (1991) is over 195,000, and the resource-based view of the firm is a hugely influential literature that forms a core part of MBA and executive education syllabi, and thus is salient to many decision makers on M&As.

³Diversifying mergers constitute approximately 50% of all mergers around the world. Although our theoretical framework of input capability applies to M&As in general, we focus on diversifying deals to avoid conflating our mechanism with other well-studied motives underlying horizontal and vertical mergers.

exemplified by real-world cases such as 3M’s acquisition in 2019 of Acelyt, a leader in advanced wound care. 3M’s core competence is its deep expertise in materials science, specifically skin-contact adhesives and advanced films. By acquiring Acelyt, whose products rely critically on specialized wound dressings, 3M could apply its specific skills to revolutionize Acelyt’s core product, thereby deploying its own capability to transform a business in a different, high-value industry.

We exploit China’s WTO accession in 2001 as a natural experiment that generated plausibly exogenous variations in input market access and input capability enhancement. Before joining the WTO, China maintained high tariffs on many intermediate inputs, creating significant barriers to the sourcing of foreign inputs. The mandatory tariff reductions following the accession to the WTO created differential shocks to the input markets of different industries (Brandt et al., 2017). Previous research demonstrates that industries experiencing larger reductions in import tariffs for their inputs tend to utilize a greater quantity, variety, and quality of these inputs.⁴ This improved accessibility to inputs enables Chinese firms to better capitalize on their input capability, which we demonstrate facilitates acquisitions of targets that use similar inputs. That is, industry pairs with significant input overlap and larger import tariff reductions are better positioned to realize merger synergies through the improved utilization of input capabilities.

We adopt a difference-in-differences (DID) approach to estimate the causal effect of trade liberalization on diversifying mergers. Specifically, following Boehm et al. (2022), we define the exposure to input-capability shocks as the degree of tariff reductions of inputs common to an industry pair. Industry pairs that shared inputs substantially and experienced significant input tariff reductions were more exposed to trade liberalization and constitute the treatment group, while industry pairs with limited input overlap or small input tariff reductions in their common inputs were less exposed to trade liberalization and constitute the control group.⁵

We control for the contemporaneous tariff reductions in common customers since mergers be-

⁴For examples, see Amity and Konings (2007); Goldberg et al. (2010); Bas and Strauss-Kahn (2015); Liu and Qiu (2016).

⁵While a growing literature studies mergers at the firm-pair level (Jovanovic and Rousseau, 2002; Rhodes-Kropf and Robinson, 2008; David, 2021), we work at the industry-pair level because: (i) the sparsity of diversifying mergers makes the industry pair a natural unit for extensive-margin analysis; (ii) reliable data on firm-level input structure is not available; and (iii) our core question—why do diversifying M&As happen between seemingly unrelated industries?—is inherently one about industry pairs, making this the most theoretically appropriate unit of analysis. We nevertheless complement our baseline with a firm–industry level analysis following Boehm et al. (2022); results reported in the Online Appendix B confirm our findings.

tween industries with similar buyers can potentially yield synergistic benefits, such as acquiring market power by integrating firms from another industry that serve similar customers (Dhingra, 2013). We also control for a rich set of fixed effects, including acquirer-industry by year, target-industry by year, and industry-pair fixed effects, which allows us to eliminate the effects of time-varying industry shocks, such as contemporary tariff reductions and other industry trends of the acquirer and target industries, as well as other time-invariant industry-pair characteristics, such as various similarities between industries that affect M&As.⁶

Our results show that industry pairs with greater exposure to the common input tariff shocks exhibit a significantly higher likelihood of diversifying mergers after the shock, compared to those that face smaller reductions in common input tariffs. This finding confirms that trade liberalization enables firms to expand into other industries that utilize similar inputs through M&As. The economic magnitude is substantial: moving the exposure degree from the 25th to the 75th percentile raises the post-WTO merger probability by approximately 0.15 percentage points, or 20% of the sample mean. A series of robustness checks, including dynamic analysis showing parallel pre-trends, confirm the validity of our research design.

To illuminate the underlying mechanism, we go beyond documenting that input similarity drives mergers and further investigate *which types of inputs* matter most. Theories of the firm suggest that when inputs are specific and require specialized knowledge, transacting at arm’s length is costly, creating an incentive to manage these activities within the firm’s boundaries (Grossman and Hart, 1986). Firms that use such inputs develop tacit expertise, such as identifying suppliers, negotiating quality, and customizing procurement: capabilities that are valuable, rare, and difficult to imitate (Barney, 1991). By contrast, homogeneous inputs traded at reference prices confer little transferable advantage, as any firm can source them efficiently on the open market. When input tariffs fall, firms with expertise in specific inputs gain access to a wider variety and higher quality of those inputs, raising the value of deploying this capability in a new industry through acquisition. We test this prediction using three complementary proxies for input specificity—the Rauch (1999) classification of differentiated goods, R&D intensity, and patent intensity—and find that the effect

⁶For example, product market overlap (Hoberg and Phillips, 2010), technological proximity (Bena and Li, 2014), human capital complementarity (Lee et al., 2018), and cultural alignment (Bereskin et al., 2018).

of common-input tariff reductions on diversifying mergers is concentrated among specific inputs. This result reveals the nature of the mechanism: the valuable capability is not about sourcing generic inputs more cheaply, but about mastering complex, knowledge-intensive ones.

We also discuss two alternative explanations: the positive sign of the main coefficient is inconsistent with the bargaining power hypothesis, under which cheaper inputs would reduce the incentive to merge for supplier leverage; and interacting the treatment with three industry-level economies of scale proxies yields uniformly insignificant terms that do not attenuate the main effect, a result that does not support the economies of scale channel as the primary driver.

Finally, we confirm the external validity of our core finding in a comprehensive global sample of diversifying mergers in 71 countries from 1998–2018, in a similar vein as [Alfaro et al. \(2016\)](#). Results demonstrate that this is a generalizable economic phenomenon.

Our work makes two primary contributions to the resource-based theory of the firm. As noted by [Chen et al. \(2022\)](#), this influential literature forms a core part of MBA and executive education syllabi and is thus salient to many decision-makers of M&As. First, we show that input capabilities are a powerful driver of *inorganic growth*, a fundamentally different corporate strategy than the *organic growth* previously studied. While [Boehm et al. \(2022\)](#) showed that input capabilities shape what firms produce internally, we demonstrate that these same capabilities motivate the far more consequential decision to change the firm’s organizational boundaries by acquiring an external company. This provides a new, capability-based rationale for the prevalence of diversifying mergers in the modern economy.⁷

Second, by studying this mechanism, we provide new evidence on the *nature of core competences themselves*. Our finding that the effect is concentrated among differentiated and innovation-intensive inputs is new to the literature and reveals what a valuable ‘input capability’ truly is: not the ability to manage simple inputs more cheaply, but the mastery of complex, knowledge-intensive ones that are difficult to replicate at arm’s length. This insight allows us to connect the abstract concept of capabilities to tangible input characteristics. Ultimately, our analysis shows that trade policy’s

⁷While recent work has highlighted various dimensions of firm similarity in driving merger decisions such as product market overlap ([Hoberg and Phillips, 2010](#)), technological proximity ([Bena and Li, 2014](#)), human capital complementarity ([Lee et al., 2018](#)) and cultural alignment ([Bereskin et al., 2018](#)), the role of input-related capabilities remains underexplored. One exception is the recent work by [Ding \(2023\)](#), who finds that various knowledge inputs improve firms’ joint production between industries in the United States.

impact is not confined to marginal production decisions; it strikes at the core of corporate strategy, altering not only what firms produce, but also how they are structured and grow.

Our paper also contributes to the trade literature by providing a new channel through which trade liberalization affects firm structure and resource reallocation. Existing research in the trade literature shows that input tariff reductions improve firm productivity through better access to foreign inputs (e.g., [Amiti and Konings, 2007](#)). We show that input tariff reductions facilitate cross-industry resource reallocation by encouraging firms to acquire targets in different industries. Although [Breinlich \(2008\)](#) and [Liu et al. \(2019\)](#) also study how trade liberalization affects resource allocations through M&A activities, they focus on horizontal and vertical M&As, respectively, whereas our paper complements them by studying diversifying mergers, which involve different industries.⁸

2 Data

Our merger analysis is based on the SDC Platinum database, which contains transaction-level information on worldwide M&A transactions involving at least 5% ownership of the target, with an at least US\$1 million or an unknown transaction value.⁹ We construct two datasets from SDC.

The first dataset, called the *Chinese merger dataset*, includes all completed domestic diversifying mergers in China between 1998 and 2007. In particular, we exclude horizontal mergers where acquirers and targets belong to the same industry, vertical mergers where acquirers and targets have a significant supplier-customer relationship,¹⁰ and mergers involving the financial industry which have special motivations. Identification of vertical mergers requires information on industry input requirements and we use the 2002 Chinese Input-output (IO) table to classify vertical mergers.

Our second dataset, called the *global merger dataset*, contains all completed domestic diversifying

⁸Our emphasis on core competences also echoes the trade literature on multiproduct firms, which currently adopts a narrower scope of “core competences” and uses it to refer to the core product that a firm can produce with the least marginal cost, such as [Bernard et al. \(2011\)](#), and [Arkolakis et al. \(2021\)](#). However, as pointed out by [Ding \(2023\)](#), such a modeling approach precludes cost or input complementarity from multi-industry operations and can be restrictive.

⁹This database is widely used in empirical work related with M&As in economics and corporate finance (e.g., [Breinlich, 2008](#); [Liu et al., 2019](#); [David, 2021](#)).

¹⁰In a vertical merger, one party (the acquirer or target) provides more than 5% of the total inputs for the production of the other ([Fan and Lang, 2000](#)). Our results are robust when the cutoff is set to 1%.

mergers of all countries in the SDC database between 1998 and 2018. We exclude cross-country mergers to avoid their complex motives, such as cross-country tax differences and incentives for technology or brand adoption, which are beyond the scope of our analysis. We use the 1997 U.S. IO table—a common approach in cross-country studies (e.g., [Alfaro et al., 2016](#))—to measure input requirements between industries for all countries. We apply the same filters as those for the Chinese merger dataset, and exclude countries with inactive merger markets that concluded fewer than 100 M&A deals during the sample period, and also exclude tax havens as well as Hong Kong and Taiwan. Ultimately, the dataset covers 71 economies from 1998 to 2018.

Table A1 and Figure A1 in the Online Appendix present the summary statistics and trends of M&As in China.

3 Empirical Design and Results

3.1 Empirical Design

As required by WTO entry, China significantly reduced import tariff rates on a wide range of intermediate inputs in 2002. These tariff cuts are widely considered exogenous in the literature. This literature documents rich evidence that industries with larger reductions in input tariffs benefit from more varieties, lower prices, and higher quality of the imported inputs in their production ([Amiti and Konings, 2007](#); [Brandt et al., 2017](#)). We explore the case in which improved access to these inputs enables firms to capitalize more effectively on their input capabilities, thereby improving their incentives to acquire targets that use similar inputs more efficiently than those with less common inputs.

First, we compute input similarity using the intermediate input matrix in our IO table, in line with [Boehm et al. \(2022\)](#). Let S be the set of all industries. The input similarity between industries i and j is defined as $IS_{ij} = \sum_{k \in S} ss_{ki} \times ss_{kj}$, where $ss_{ki} = \frac{X_{ki}}{\sum_{h \in S} X_{hi}}$ is the input share of k in industry i , where X_{hi} denotes the value of output from industry h used in the production of industry i . ss_{ki} measures the importance of input k among all inputs of industry i . ss_{kj} is similarly defined for industry j . In particular, IS_{ij} is greater when industries i and j exhibit a higher degree of input

overlap.

Next, we introduce a measure of exposure to the tariff shocks in input capability. We measure exposure using the degree of tariff reductions in common inputs within each pair of industries, following the method of [Boehm et al. \(2022\)](#). The rationale is that industry pairs with significant input overlap and larger tariff reductions are better positioned to realize merger synergies through improved utilization of input capabilities. Specifically, we have

$$IS-T01_{ij} = \sum_{k \in S} \tau_{k,2001} \times ss_{ki} \times ss_{kj}, \quad (1)$$

where $\tau_{k,2001}$ is the log of the average import tariff level of industry k in year 2001, before the tariff cuts necessitated by WTO accession. In particular, we follow the literature to use an industry's initial tariff levels to measure actual tariff reductions in China (e.g., [Amiti and Konings, 2007](#)). Initial tariff levels are highly positively correlated with actual tariff reductions and remained stable for years before China entered the WTO, making them a plausibly exogenous variable ([Liu et al., 2019](#)). By incorporating the predetermined level of input usage structure and tariff rates in 2001, we can classify industry pairs with higher (lower) values of $IS-T01_{ij}$ as the treatment (control) group.

We conduct a DID analysis to identify the causal effect. Specifically, we estimate the following specification with the Chinese merger data:

$$MA_{ijt} = \gamma \cdot IS-T01_{ij} \times Post_t + \alpha_{ij} + \alpha_{it} + \alpha_{jt} + \epsilon_{ijt}, \quad (2)$$

where MA_{ijt} is a dummy variable indicating whether there is at least one diversifying merger between a firm of industry i and another firm of industry j in year t . $IS-T01_{ij}$ is the intensity of import tariff shock to industry pairs with similar inputs defined above and $Post_t$ is a dummy variable that equals 1 for the year 2002 and onwards, and 0 otherwise. We include both acquirer-industry-year (α_{it}) and target-industry-year (α_{jt}) fixed effects to control for time-varying industry-specific factors driving merger waves. Since we rely on time-varying shocks at the industry-pair level for identification, we can further include the industry-pair fixed effects α_{ij} to control for all time-

invariant factors at the industry-pair level, such as the co-location of industries with similar inputs, and various dimensions of similarity between industries such as technology and human capital similarity. This approach significantly strengthens our ability to isolate the causal relationship by effectively addressing potential unobserved confounding factors that may exist at the industry-pair level. We cluster standard errors at the industry-pair level (Abadie et al., 2023).

We conjecture that $\hat{\gamma}$ is positive as industry pairs that share more common inputs with larger tariff reductions are more likely to have mergers between them.

3.2 Baseline Results

Column (1) of Table 1 presents the regression results of Eq. (2). The results indicate that when two industries exhibit higher initial levels of input similarity and experience larger reductions in import tariffs of their shared inputs, their firms are significantly more likely to have diversifying mergers.

Mergers between industries with similar customers can also yield synergistic benefits, such as acquiring market power by integrating firms from another industry that serve similar customers (Dhingra, 2013). China’s accession to the WTO brings contemporaneous tariff shocks to the common customers of each industry pair, and thus may influence mergers between industries. To account for this potential effect, we first define output similarity as $OS_{ij} = \sum_{k \in S} bs_{ik}bs_{jk}$, where bs_{ik} denotes the buyers’ share of industry k among all buyers of product i . Then analogously to tariff shocks to common inputs, we include $OS-Tariff_{ijt} = (\sum_{k \in S} \tau_{k,2001} \times bs_{ik} \times bs_{jk}) \times Post_t$ to control for how tariff reductions affect the mergers between industries of common customers. Column (2) shows that our coefficient of interest ($\hat{\gamma}$) remains significant at the 1% level.

The economic significance of tariff shocks to common inputs is notable. Based on the estimate in Column (2), an increase in $IS-Tol_{ij}$ from the 25th to the 75th percentile corresponds to a 0.15 percentage point increase in the probability of a merger occurring between industries i and j . This effect represents approximately 20% of the average merger propensity across industry pairs (0.72 percentage points), indicating that input similarity substantially affects the likelihood of diversifying mergers.

In addition to the extensive margins, we also examine how input similarity affects the intensive

margins of mergers. We redefine the outcome variable (MA_{ijt}) in Eq. (2) as the count of mergers between industries i and j in year t , denoted by N_{ijt} . We rerun the regression and report the results in Column (3). The key estimate remains positive and statistically significant, indicating that industries with higher input similarity tend to have a greater number of mergers. This effect is largely driven by the highly sparse distribution of cross-industry M&A activities on an annual basis. Among industry pairs with at least one merger in a year, 82% of them have only one merger, and 13% have two. Hence, although the results regarding the intensive margins are consistent with those of the extensive margins, we focus on the extensive margins in the rest of the paper.¹¹

A critical assumption for our identification strategy is that without the differential changes in input tariffs caused by China’s WTO entry, there would be no differential changes in merger activities across industry pairs. Although we cannot directly test this counterfactual assumption, we can provide corroborative evidence by examining whether the treated and control industry pairs have common pre-trends in M&A before the WTO entry. We replace $Post_t$ with a vector of year dummies and estimate the following flexible specification:

$$MA_{ijt} = \sum_{t=1999}^{2007} \gamma_t \cdot year_t \times IS-T01_{ij} + OS-Tariff_{ijt} + \alpha_{ij} + \alpha_{it} + \alpha_{jt} + \epsilon_{ijt}. \quad (3)$$

The time-specific variable for the year 1998 is the omitted baseline. This specification allows the merger propensity to vary flexibly with time and input similarity.

Figure 1 plots the results. The coefficients for the years before China’s WTO entry are statistically insignificant and approximately zero, suggesting that there are no differential trends in merger probability between the treated and control industry pairs prior to the significant tariff shocks. However, in the post-WTO years, the coefficients become positive and statistically significant, indicating a causal impact resulting from the tariff reductions. These findings further support the validity of our identification strategy.

To ensure the robustness of our findings, we conduct a battery of checks to investigate the sensitivity of our results to various specifications and potential confounding factors. First, we complement the baseline industry-pair analysis with a firm–industry panel following [Boehm et al.](#)

¹¹Estimates obtained using the Poisson Pseudo Maximum Likelihood estimator remain robust.

(2022) and show that the results are unchanged when acquirer-firm-by-year fixed effects absorb all time-varying firm-level heterogeneity (see Table ?? in the Online Appendix). Second, we control for the contemporaneous reduction in US trade policy uncertainty using both a Column 2 tariff shock and a continuous tariff-gap proxy; neither uncertainty measure is significant, and our main coefficient is unaffected (Table ??). Finally, we conduct a battery of additional checks that include defining vertical mergers using an alternative cutoff (1% instead of 5%) for vertical relationships; defining horizontal mergers based on a coarser sector level; using cosine similarity to measure input similarity; replacing the initial-year tariff shock with time-varying tariffs; dropping the top 5% of industry pairs ranked by input similarity; restricting the sample to manufacturing industries; and using robust standard errors clustered two-way at the acquirer- and target-industry levels. Our results hold in all of these tests (Table ??).

3.3 Mechanism

Having established the baseline effect, we now examine which types of common inputs drive it. If the mechanism operates through input capability—tacit expertise in sourcing and utilizing specific inputs—then the effect should be concentrated in industry pairs whose shared inputs are harder to obtain through arm’s-length transactions. Specific inputs, such as differentiated intermediates, R&D-intensive components, and patent-intensive goods, require relationship-specific investments in supplier identification, quality assurance, and procurement customization (Grossman and Hart, 1986; Nunn, 2007; Barrot and Sauvagnat, 2016). The expertise accumulated through these activities constitutes a transferable core competence: when tariffs on such inputs fall, the value of deploying this know-how in a new industry through acquisition rises. By contrast, homogeneous inputs traded at reference prices can be sourced efficiently by any firm, so shared usage confers little advantage that would motivate a merger. We test this prediction by interacting the baseline treatment with three established measures of input specificity and examining whether the merger response to tariff reductions varies accordingly.¹²

¹²We confirm that cross-sectional variations in import tariff reductions are orthogonal to input specificity (see Tables ?? and ?? in the Online Appendix). This enables us to isolate the effects of input specificity conditional on tariff reduction levels.

3.3.1 Differentiated Inputs

Differentiated inputs are more likely to be specific and require relationship-specific investments, while homogeneous inputs tend to be non-specific (Nunn, 2007). Accordingly, we examine whether tariff shocks to differentiated common inputs have a greater impact on determining M&As than homogeneous common inputs. We define the level of product differentiation of industry k , $shDF_k$, as the share of the constituent HS product codes that are classified as differentiated according to Rauch (1999), and the tariff shock on differentiated common inputs as $IS-T01-shDF_{ij} = \sum_k shDF_k \times \tau_{k,2001} \times ss_{ki} \times ss_{kj}$. We use this new measure to estimate the following specification:

$$MA_{ijt} = \gamma_1 \cdot IS-T01_{ij} \times Post_t + \gamma_2 \cdot IS-T01-shDF_{ij} \times Post_t + \alpha_{ij} + \alpha_{it} + \alpha_{jt} + \epsilon_{ijt}. \quad (4)$$

The regression results are presented in Column (1) of Table 2. $\hat{\gamma}_2$ is positive and statistically significant at the 1% level, which indicates that tariff reductions have a significantly greater impact on mergers when common inputs are more differentiated.

3.3.2 Innovation-intensive Inputs

Following Barrot and Sauvagnat (2016), we use innovation intensity to proxy input specificity. We argue that firms that use inputs with high innovation intensity are likely to possess core competences in their usage, as the embedded knowledge is built on innovation activities. We expect that trade liberalization will have a greater impact on M&As in industry pairs that share inputs with a higher intensity of innovation.

We measure innovation intensity using two approaches: patent intensity and R&D intensity. We first calculate the patent intensity ($shPT_k$) and show that the patent-intensity interaction is positive and statistically significant at the 5% level (Column 2), confirming that the effect is more pronounced when common inputs are more patent-intensive. We then use the average share of R&D expenses over sales ($shRD_k$) and construct $IS-T01-shRD_{ij} = (\sum_{k \in S} shRD_k \times \tau_{k,2001} \times ss_{ki} \times ss_{kj})$ to capture tariff shocks on R&D-intensive common inputs. Re-estimating Eq. (4) with $IS-T01-shRD_{ij}$, the coefficient on the R&D-intensity interaction is positive though not statistically

significant (Column 3).

3.3.3 Discussions on Alternative Explanations

Our identification strategy leverages variations in import tariff reductions on common inputs between industries, which ensures that other confounding factors do not undermine our findings. However, two alternative explanations could potentially predict the relationship between input trade liberalization and mergers.

The first alternative explanation is the buyer power hypothesis, which posits that firms with common suppliers might merge to increase their bargaining leverage in sourcing markets (e.g., [Horn and Wolinsky, 1988](#)). While this is a plausible motive for mergers in general, the context of China’s WTO accession allows us to distinguish this mechanism from our proposed input capability channel, as they generate opposing predictions.

Specifically, if the primary motive for merging were to gain buyer power, a large reduction in input tariffs would weaken this incentive. The trade liberalization provided firms with greater access to cheaper and more diverse inputs on the global market, thereby improving their outside options and strengthening their bargaining position with domestic suppliers even without merging. Consequently, the buyer power hypothesis predicts that larger common-input tariff reductions should lead to fewer diversifying mergers.

In contrast, our input capability mechanism predicts the opposite. For firms with shared sourcing competences, lower input tariffs reduce the cost of leveraging this know-how, making it more attractive to acquire a target with a similar input structure to create synergies. This predicts that larger tariff reductions should lead to more diversifying mergers. Our central empirical finding—a robust positive coefficient on the interaction term—is therefore consistent with the input capability channel and inconsistent with a motive primarily driven by enhancing buyer power. While we cannot rule out that firms consider bargaining power, our results suggest that in the context of this major trade liberalization, the incentive to leverage sourcing competences was the dominant force driving these merger decisions.

The second alternative explanation is economies of scale rather than bargaining power. Under this view, firms merge to exploit increasing returns by spreading fixed operating costs or jointly fi-

nancing R&D and innovation in input sourcing, regardless of whether the shared inputs are specific or homogeneous. To assess this mechanism, we interact our tariff-based common-input measure with three industry-level proxies for scale economies in the acquirer industry: the scale-elasticity estimates of [Antweiler and Trefler \(2002\)](#), the share of nonproduction workers in total employment following the proximity-concentration literature (e.g., [Brainard, 1997](#)), and the share of workers with at least a college education. These measures capture complementary dimensions of scalability, including increasing returns in production, the importance of firm-level fixed costs such as management and R&D, and the intensity of skilled scalable tasks. If scale economies were the main mechanism, these interaction terms should be large and significant and should attenuate the coefficient on our main variable. Instead, across all three specifications, the interaction terms are statistically weak and the coefficient on our main variable remains positive and significant (Table A7 in Appendix). These results suggest that economies of scale are unlikely to be the primary channel, and instead support our interpretation that the value of diversifying M&As reflects capabilities tied to specific inputs.

4 Global Evidence

We have used the institutional setting and granular data from China to show the causal evidence and explore the mechanisms. In this section, we use our global dataset of diversifying mergers to generalize our findings to 71 countries during 1998-2018, in a similar spirit as [Alfaro et al. \(2016\)](#).

Analogously to the China case, we run the following regression:

$$MA_{cijt} = \beta \cdot IS-Tariff_{ijt}^c + \alpha_{ij} + \alpha_{it} + \alpha_{jt} + \alpha_{ct} + \epsilon_{cijt}, \quad (5)$$

where $IS-Tariff_{ijt}^c = (\sum_{k \in S} \tau_{k,t}^c \times ss_{ki}^{US} \times ss_{kj}^{US})$. We use the US IO table to measure input requirements between industries for all countries to ensure consistency in input classifications. In particular, ss_{ki}^{US} is the input share of k in industry i from the US IO table. $\tau_{k,t}^c$ is the log average applied tariff in year t in country c industry k . To accommodate time-varying country characteristics, we further control for country-year fixed effects α_{ct} . Robust standard errors are clustered

at the industry pair level. We also construct the $OS-Tariff_{ijt}^c = (\sum_{k \in S} \tau_{k,t}^c \times bs_{ki}^{US} \times bs_{kj}^{US})$, with bs_{ki}^{US} denoting buyer k 's share in industry i from the US IO table. This variable is also included to control for the tariff's impacts through the buyers' similarity. Note that the tariffs are included directly instead of using the initial year level as in the case of Chinese data. Therefore, we expect $\hat{\beta}$ to be negative.

The regression results are reported in Table 3. Column (1) uses the full global sample and includes only $IS-Tariff_{ijt}^c$. Column (2) adds the buyer-side tariff measure $OS-Tariff_{ijt}^c$. Columns (3) and (4) retain both tariff-based channels and split the sample by income group, with Column (3) for higher-income economies and Column (4) for lower-income economies. Across Columns (1) and (2), lower common-input tariffs are associated with a higher likelihood of mergers between industry pairs in the global data. The effect remains positive and significant in the higher-income subsample in column (3), and positive though statistically insignificant in the lower-income subsample.¹³ These results confirm that our main findings are general, not confined to developing economies or China's unique institutional settings.

5 Conclusion

We draw on insights from the resource-based theory of the firm (Penrose, 1955) to demonstrate that input capability drives diversifying mergers. Our study reveals that firms are more likely to engage in M&As to diversify into industries with similar input structures, to leverage their input capability. To establish causality, we use exogenous import tariff reductions from China's accession to the WTO as a natural experiment. Our analysis indicates that industry pairs with similar input usage before tariff reductions are more likely to merge if their common inputs are subject to larger tariff reductions. This effect is stronger for differentiated and innovation-intensive inputs, consistent with the input capability channel. These results suggest that trade liberalization impacts aggregate productivity by influencing M&As and resource allocation. The basic finding obtained using Chinese data is also supported by evidence from the global sample, which includes 71 countries.

¹³We interpret the weaker result for lower-income economies cautiously: the incidence of diversifying M&As in these economies (0.23%) is substantially lower than in higher-income economies (0.47%), which limits variation in the dependent variable and reduces statistical power.

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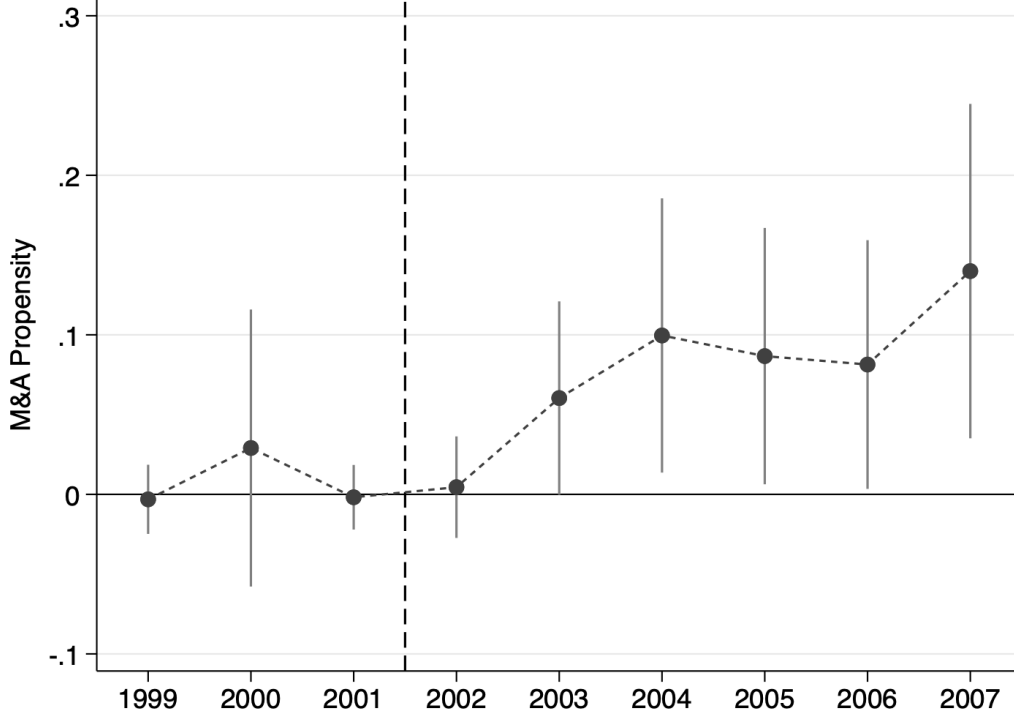
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Tables and Figures

Figure 1: Flexible Estimation



Note: This figure plots the estimated coefficients $\hat{\gamma}_t$ and their 95% confidence intervals from the flexible specification $MA_{ijt} = \sum_{t=1999}^{2007} \gamma_t \cdot year_t \times IS-T01_{ij} + OS-Tariff_{ijt} + \alpha_{ij} + \alpha_{it} + \alpha_{jt} + \varepsilon_{ijt}$, estimated on the Chinese merger sample during 1998–2007. The dependent variable MA_{ijt} is an indicator for whether at least one diversifying merger occurs between industries i and j in year t . The key independent variable $IS-T01_{ij} = \sum_{k \in S} \tau_{k,2001} \times ss_{ki} \times ss_{kj}$ captures the intensity of the pre-WTO (year 2001) input tariff shock to industry pairs with similar inputs. Year 1998 is omitted as the reference year. We control for the tariff reduction on common buyers, $OS-Tariff_{ijt}$, and include industry-pair fixed effects α_{ij} , acquirer-industry-by-year fixed effects α_{it} , and target-industry-by-year fixed effects α_{jt} . Standard errors are clustered at the industry-pair level.

Table 1: Baseline Results

	MA_{ijt}		N_{ijt}
	(1)	(2)	(3)
$IS-T01_{ij} \times Post_t$	0.0819***	0.0727***	0.1120***
	(0.0164)	(0.0163)	(0.0342)
$OS-Tariff_{ijt}$		0.0393**	0.0475
		(0.0200)	(0.0302)
Fixed Effects		ij, it, jt	
Observations	138,050	138,050	138,050

Note: The dependent variable in columns (1)–(2) is MA_{ijt} , an indicator of whether industry i and industry j have any diversifying merger in year t . In column (3) the dependent variable is N_{ijt} , the number of such mergers. $IS-T01_{ij} \times Post_t$ measures the common-input tariff shock, and $OS-Tariff_{ijt}$ is the analogous buyer-side tariff-shock control. Standard errors clustered at the industry-pair level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2: Mechanism

	MA_{ijt}		
	(1)	(2)	(3)
$IS-T01_{ij} \times Post_t$	-0.0268 (0.0340)	0.0169 (0.0237)	0.0522** (0.0243)
$IS-T01-shDF_{ij} \times Post_t$	0.1065*** (0.0406)		
$IS-T01-shPT_{ij} \times Post_t$		0.0700** (0.0296)	
$IS-T01-shRD_{ij} \times Post_t$			0.1027 (0.0955)
Controls		$OS-Tariff_{ijt}$	
Fixed Effects		ij, it, jt	
Observations	138,050	138,050	138,050

Note: The dependent variable is MA_{ijt} , an indicator of whether industry i and industry j have any diversifying merger in year t . The three interaction terms capture common-input tariff shocks weighted by differentiated-input share, patent intensity, and R&D intensity, respectively. All regressions include the buyer-side tariff-shock control $OS-Tariff_{ijt}$. Standard errors clustered at the industry-pair level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3: Global Evidence

	MA_{ijt}^c			
	(1) Global	(2) Global	(3) Higher Income	(4) Lower Income
$IS-Tariff_{ijt}^c$	-0.0535*** (0.0100)	-0.0440*** (0.0097)	-0.0524*** (0.0105)	0.0060 (0.0045)
$OS-Tariff_{ijt}^c$		-0.0518*** (0.0109)	-0.0816*** (0.0144)	0.0270* (0.0143)
Fixed Effects	it, jt, ij, ct			
Observations	4,716,033	4,716,033	3,121,881	1,594,152

Note: The dependent variable is MA_{ijt}^c , an indicator of whether industry i and industry j in country c have any diversifying merger in year t . Column (1) includes only the common-input tariff measure. Columns (2)–(4) additionally control for the buyer-side tariff measure. Standard errors clustered at the industry-pair level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.